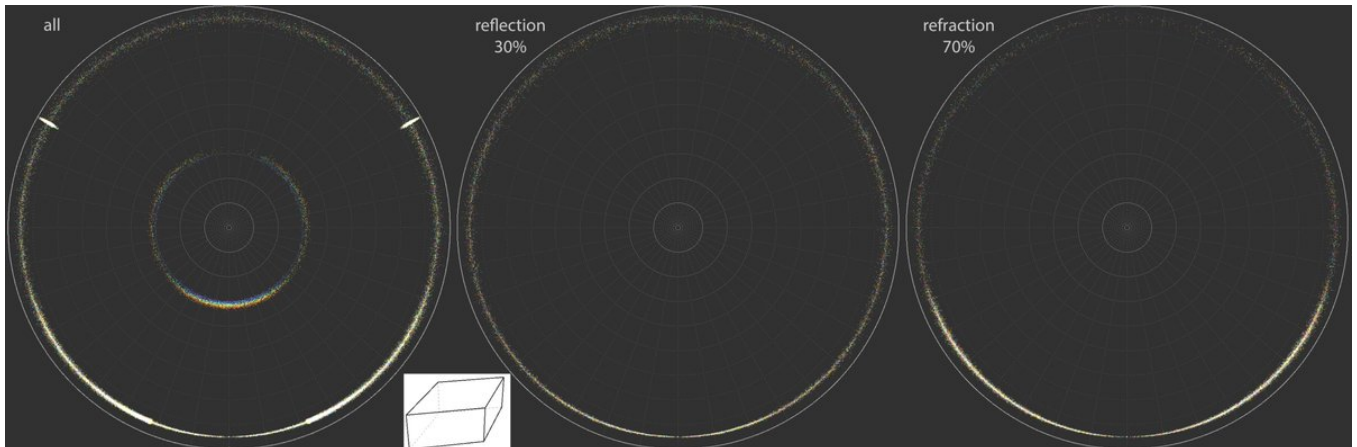


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<https://x.com/RiikonenMarko/status/1959226712371118440>

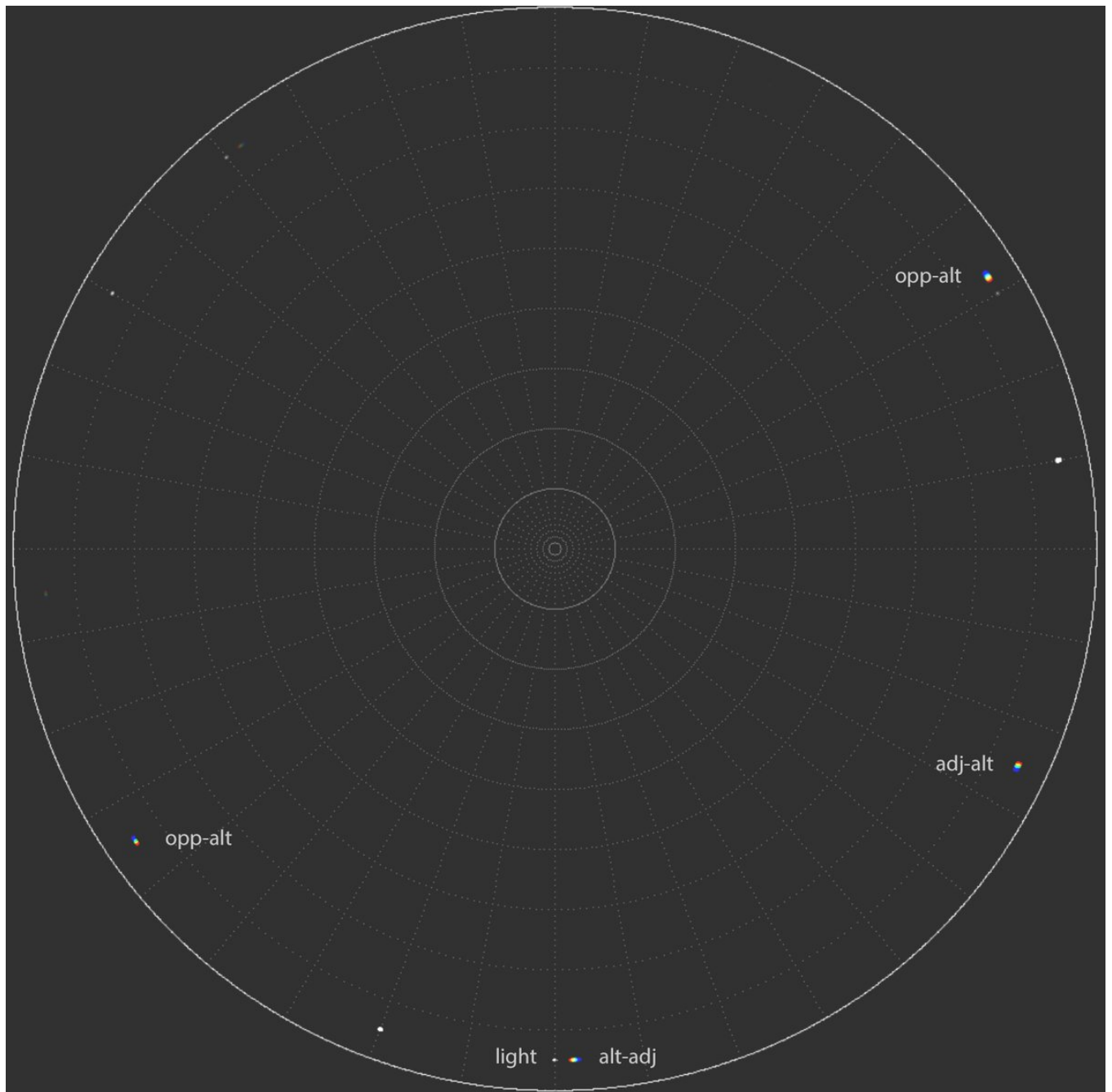
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Parhelic circle is regarded as a reflection halo. But suppose there was a population of plate oriented diamonds. Then, at very low light elevations, the parhelic circle is dominated by refraction raypaths. These come in 4 variations, 356, 346, 358, 368, and might be coded more



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generally as alt-adj, adj-alt, alt-opp, opp-alt. In rotating crystals, the raypaths make white segment, az-locking transforms them into the colored parhelion spots. The shown HaloPoint simulation with az-lock value 130° turns up three of these triple-hit spot types. The other



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spots are those of reflection parhelic circle (contributed, in the order of dominance, by raypaths 3, 132 and 363), 120° parhelia and 120° rotated parhelia (weakly at 142° azimuth on the left). At the simulation's light elevation of 5 degrees the refraction parhelic circle

HALOPOINT 2.0

☒ Save work-file
☐ Save light work-file
☒ Open in Analysis Window

Start Simulation

Add to the Simulation

General Parameters

Sun Elevation [°]
Sun Diameter
Number of rays
Colour of Light:
☐ Single Wavelength [nm]
☒ Wavelength Range
From nm to nm
Number of hits allowed:

Simulation View

Lens Type
Focal Length [mm]
Sensor Size [mm] W: H:
☐ Custom parameters a: b:
Center of Image
Azimuth:
Elevation:
☐ Portrait ☒ Landscape
Rotation (CW)

Appearance

Dots
Background
Shade Levels
☒ Show Coordinate Grid
Major Step [°]
Minor Step [°]
Style
☐ Horizon ☐ Hide Subhorizon

Ice Crystal Population Parameters

	%	C - axis			Rotation			Prism Aspect Ratio			Upper Pyramid				Lower Pyramid			
		Mean	Std	Distribution	Mean	Std	Distribution	Mean	Std	Distribution	Apex	h	Std	Distribution	Apex	h	Std	Distribution
☐ 1	100	90	0	Gaussian	90	360	Gaussian	0.8	0.5	Uniform	56,14	0	0	Uniform	56,14	0	0	Uniform
☐ 2	400	90	0.3	Gaussian	90	0.3	Gaussian	1	0.5	Uniform	56,14	0	0	Uniform	56,14	0	0	Gaussian
☐ 3	100	90	0.3	Gaussian	90	0.3	Gaussian	0.8	0.4	Uniform	56,14	0	0	Uniform	56,14	0	0	Gaussian
☐ 4	100	90	0.3	Gaussian	90	360	Gaussian	0.8	0.5	Uniform	56,14	0	0	Gaussian	56,142	0	0	Gaussian
☒ 5	800	0	0.1	Gaussian	90	0	Uniform	0.3	0	Uniform	39,2	0	0	Gaussian	56,14	0	0	Gaussian
☐ 6	50	0	1	Gaussian	90	360	Gaussian	0.8	0.4	Uniform	39,2	0	0	Gaussian	39,2	0	0	Gaussian
☐ 7	200	0	1	Gaussian	90	360	Gaussian	0.7	0.1	Uniform	56,14	0	0	Uniform	56,14	0	0	Gaussian
☐ 8	40	90	0	Gaussian	90	360	Uniform	0.7	0.3	Uniform	39,2	0	0	Gaussian	39,2	0	0	Gaussian
☐ 9	120	90	0	Gaussian	90	360	Uniform	2	0.3	Uniform	56,142	0	0	Gaussian	56,142	0	0	Gaussian
☐ 10	400	90	0	Gaussian	90	0	Gaussian	3	0.5	Uniform	56,142	0	0	Gaussian	56,142	0	0	Gaussian

Visualisation

Prism Face Distances

Apply	Face Distance from c-axis								Std of Distance								Sync	Distribution
	3	4	5	6	7	8	3	4	5	6	7	8						
☐ 1	2	1	2	1	2	1	0	0	0	0	0	0	0	0	Uniform			
☒ 2	1	2	1	1	2	1	0	0	0	0	0	0	0	0	Uniform			
☒ 3	1	1	2	1	1	2	0	0	0	0	0	0	0	0	Uniform			
☒ 4	1	2	1	1	2	1	0	0	0	0	0	0	0	0	Uniform			
☒ 5	1	2	1	1	2	1	0	0	0	0	0	0	0	0	Uniform			
☒ 6	2	1	2	1	2	1	0.3	0.3	0.3	0.3	0.3	0.3	0	0	Uniform			
☐ 7	1	2	1	1	2	1	0	0	0	0	0	0	0	0	Uniform			
☒ 8	1,2	1	1,2	1	1,2	1	0.2	0	0.2	0	0.2	0	0	0	Uniform			
☒ 9	1,2	1	1,2	1	1,2	1	0.2	0	0.2	0	0.2	0	0	0	Uniform			
☒ 10	2	1	1	2	1	1	0	0.3	0	0	0.3	0	0	0	Uniform			

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raypaths dominate at $h/d > 0.09$. At higher light, reflection raypaths gain upper hand. Maybe we should recognize the triple-hit type as e.g. "free rotated (refraction) parhelia", as opposed to the "120° rotated parhelia". In Herincs displays we seem to have both types observed.